MODULE 8

Activity submission scenario

UNIVERSITY OF CAPE TOWN

BUSINESS SYSTEMS ANALYSIS

IN COLLABORATION WITH getsmarter

Learning outcome:

LO3: Forumlate data definitions.

Instructions

This scenario expands on the Kabeljou scenario introduced in previous modules. All information presented previously and in this scenario document is to be considered for the activity submission in this module. Quiz questions will not be based on this scenario, unless stated otherwise.

Note:

This scenario is entirely fictitious. Although many details are inspired by real life, resemblance to any real-life person or entity or event is purely coincidental and unintentional.

Scenario

So far, your various models and diagrams and other specifications have helped Kabeljou's e-commerce project achieve rapid progress. You're getting complimentary feedback from your project sponsor and other colleagues. It's especially pleasing to get Vuyelwa's praise because, compared with "regular" business users, her role as project manager and her IT background make her a lot more knowledgeable in many of the tools you've introduced to the business.

It's now time to get right down to what is often considered the core of any computer system: specifying the data requirements. In preparation for a workshop with Vuyelwa and Megan (the Sell It specialist contractor), you need to prepare portions of the main data modelling tools, namely, the entity relationship diagram (ERD), data flow diagram (DFD), and some of the data definitions of key entities.

Similar to your approach to developing the system architecture model (SAM) and functional decomposition diagram (FDD), you've scanned through your consulting notes to extract the key details that will be your input to these diagrams and documents.

Entity relationship diagram

1. The entities and their relationships you need to map for your upcoming workshop with Megan and Vuyelwa are:

    1.1. CUSTOMER

    1.2. ORDER

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1.3. STOCK_ITEM

1.4. CAMPAIGN

2. From your consulting notes, you know the following about the relationships between these entities:

   2.1. A customer registered in the system can place “none” or “many” orders, while an order can be placed by only one customer.

   2.2. An order can have zero, one or many stock items. It might be unexpected to consider that an order could exist without a stock item, but we are reminded that the cart exists in the system as an order record and it is possible for a cart to be “emptied” by a customer. A stock item can exist in many orders. Therefore, the relationship between order and stock item is many to many. Megan, who will implement the database, advised that this will need to be normalised using a logical entity named “ORDER-STOCK_ITEM”.

   2.3. A campaign can have many orders or might not be used in any orders, while an order can belong to one campaign or none at all.

It’s now time to complete the ERD that shows these entities. In the workshop with Vuyelwa and Megan, we will use the ERD as a starting point to analyse the attributes needed for each entity. So, although we would typically start with a conceptual ERD, it will be useful to develop the ERD a half a step further towards being a logical-level ERD. However, for this “discussion starter” ERD, you’ve agreed to **prepare the ERD showing normalised relationships as at the logical level, but for simplicity in the diagram, it will omit all attributes**. In other words, you need to show only the name of each entity inside its element in the ERD and not its attributes.

Being a logical-level ERD, it must show the fully-normalised entities and the cardinality of relationships between the entities using the crow’s foot notation. As a new tool to your peers at Kabeljou, it will be useful to label each relationship in both directions, i.e. from the perspective of each entity in the relationship, as described, for example, in 2.1 above where a customer *can place* orders.

Data flow diagram

A critical part of the project’s scope of work is to migrate data from the current, old systems into the new system. Because we currently don’t have a fully functioning e-commerce and CRM system, the source data is in several disparate and archaic sources, such as Excel spreadsheets. This data will need to be selectively extracted, cleaned and further transformed in preparation for importing it into the Sell It database. The overall effort is conveniently known as ETL – “extract, transform and load”.

However, because of how data will flow through the envisaged system, there is a specific sequence in which the old system’s data will need to be imported. For example, as we already

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know, data in an order record needs customer data from the customer record i.e. data flows from 'CUSTOMER' to 'ORDER'. We can, therefore, conclude that a customer's record must exist before that customer can create an order. (We've analysed this detail as part of our specification of the "Check out cart" use case.)

At a data level, the mechanics of this flow rely on the order record using the customer record's ID, its primary key, as a means of associating the customer data with the order. Specifically, the customer ID will be a foreign key in the order record.

Therefore, if we want to import order records into the new system, we must first import customer records. This is one of the practical implications of understanding data flows in a system, which you will need to model in a data flow diagram (DFD).

With Megan's help again, you have analysed the data flow requirements of the system and summarised your notes. The data flow has been analysed between the various modules within Sell It and between the modules and external systems. We have not yet analysed the data flows between data stores and processes that may exist within each module of Sell It. In other words, our DFD will be a high-level view and not detailed for now.

3. Notes for the DFD

    3.1. When a customer checks out a cart, the Order module will need a pre-existing record in the Customer module so that customer's data can automatically populate customer fields in the order, such as their name, email and delivery address. This not only saves the customer time and convenience, but it will also reduce data errors, which is intended to improve service through reducing the rate of lost deliveries. (Recall that a reduction in data capturing errors is one of the benefits analysed in the business case.)

    3.2. The data flow between Order and Customer modules is bi-directional: with reference to the "Check out cart" use case, recall that the customer's data in the order always gets saved back to the customer record. This is to ensure that any updates the customer makes to his or her information in the cart is kept for possible future use and that we always have the customer's latest information.

    3.3. The Order module, which stores each cart (AKA each order record) gets data from the Catalogue module, where stock records are stored.

    3.4. Orders will also track which campaign sparked the sale, if any. If the order is placed as a result of a campaign lead, like an email newsletter or a banner ad, then the 'ORDER' record will store the campaign ID as one of the tracking codes. Data, therefore, flows from the Campaign module to the Order module.

    3.5. Each bank's gateway information will be stored in the Payment module. When a user clicks "Pay now" in the cart checkout page, the Order module will send the billing

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data, such as the credit card details and transaction value, to the Payment module so that payment can be collected.

3.6. The Payment module will then request payment from the appropriate bank's gateway, a physical system outside of Sell It. This data flow is bi-directional: the gateway needs to send a responding message of the request being honoured or declined, which will be received by the Payment module.

3.7. During this time, the customer using the cart is still waiting for a response from the system, so the Payment module needs to send a message back to the Order module with the gateway's response.

3.8. After successful payment, Sell It's Order module sends a message to the external email system. This message must trigger emails to the customer and to the dispatch team.

3.9. Lastly, the Reporting module draws on data from the Customer, Order, and Catalogue modules in order to produce reports like sales values over a specified period, abandoned carts, customer demographics, and stock movements.

Now, to help Vuyelwa and Megan with the planning for the ETL, you need to create a DFD that shows how data moves between modules in Sell It, as well as between Sell It and to and from external systems.

Similar to the logical-level ERD without attributes, this initial DFD will be a discussion starter for you, Vuyelwa and Megan to explore the ETL requirements. As such, we want the diagram to be as simple as possible: Sell It's modules are to be treated as an aggregation of their respective processes and data stores. In other words, your high-level DFD will show only the modules. For example, although 'CUSTOMER' is a data store and "Update customer data" is a process within the Customer module, the DFD must show only the module and not the data store or process inside the Customer module. If we were to model each individual process and data store, the diagram would be too detailed and distracting for the purpose of introducing the DFD as a tool to Kabeljou.

Human actors also need not be shown. We want to emphasise and analyse only the automated data flow between the system's modules and external *systems*. For clarity of communication and specifications, it will help to show the boundary between the system's modules and external systems.

Data definitions dictionary

Another action you've committed to taking in preparation for the workshop with Vuyelwa and Megan is to draft what you think will be the specifications for some of the attributes for the 'CUSTOMER' and 'ORDER' entities. This will be similar to the content of the physical ERD, but will take the form of the start of a data definitions dictionary. For each of the two entities, you

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will need to create a table that sets out the specifications for each attribute's name, key, data type, and data size.

4. For Megan's purposes and, because of the terminology used by the Sell It system, we need to categorise each attribute as one of six different data types:

   4.1. **System**: a field that only the database system itself can write to; for example, the Sell It program can read this field, but it can't update this type of field;

   4.2. **Varchar**: variable character, used for free-format text fields;

   4.3. **Char**: character, used for menu fields with menu options that are fixed in their values;

   4.4. **Int**: integer, a field type for numbers with a precision of zero decimals;

   4.5. **Num**: number that includes decimals with a definable precision of decimals;

   4.6. **Date**: date-time fields have a fixed structure in Sell It, so it is not necessary to define a data size for this data type.

When specifying the data size for system, varchar, and char data types, we will need to state only the maximum number of characters. For number and integer types, however, we should indicate the minimum and maximum values as a range of values each field of this type is allowed. Further, for num fields, it will be useful to indicate the precision available, in other words, the number of decimals needed.

There have been no workshops with users to discuss the detailed data requirements specifically. All input you have on the data definitions thus far has been acquired incidentally from workshops for other purposes. This means that you have no reliable input to this effort in defining the low-level data requirements.

From a project perspective, by collating all the information you've elicited incidentally into a single document for your workshop with Megan and Vuyelwa, you will help improve the project's efficiency and schedule, while applying tools that create precise and detailed specifications for the intended solution. In other words, you'll be fulfilling the role of a competent BSA by helping to implement a solution that is cheaper, delivered quicker, and of higher quality than how things might have been at Kabeljou if a BSA – you – were not hired.

You had agreed with your colleagues on the fields you would analyse for your upcoming workshop. These are listed in the tables in the activity submission file. Your task now is to specify the data definitions in the empty, unshaded cells in these tables.

User experience

In your most recent, informal discussion with Marius over lunch the other day, you noticed another trait in Marius that contributes to his interpersonal and business success. Beyond the

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obvious factor of his personable, relaxed leadership style, his frequent, impromptu chats with his colleagues and his seemingly all-the-time-in-the-world way of giving attention, he also embodies something else that is more critical to online business success: Marius has empathy.

Your “canteen conversations” are clearly not a planned, formal requirement elicitation workshop, but through them you have gained valuable insight into Marius’ skills in empathy and listening. More importantly for your project, you feel inspired by how he makes a concerted effort to imagine himself in Kabeljou’s customers’ shoes. Marius asked several pointed questions about exactly how it would feel to use the e-commerce system as a customer.

He commented on some of the features he’d seen when Megan demonstrated the standard, un-customised version of Sell It to us in a recent workshop. Many of Marius’ questions during your impromptu lunch “workshop” showed his keen attention to the seemingly insignificant and often overlooked details that can make or break a system from an end user’s perspective.

You introduced Marius to the concept of personas as a way of gaining insight into who the users are. Because of Marius’ hands-on and affable style in interacting with local fishermen and customers, he shared some useful stories of Kabeljou’s typical customers. He says he had, in fact, informally interviewed a few customers after the e-commerce project started to get a personal feel for what they might like from the system.

Marius obviously has good insight into who Kabeljou’s customers are and what might be useful requirements to enhance the customer experience. He confirmed that many of these requirements can be traced back to conversations he had with customers. You remember that when UX requirements can be traced back to actual end users, they are generally more reliable than those hypothesised by system owners or biased by internal users.

Unfortunately, because of the spontaneous nature of the canteen chat, you didn’t have your notebook with you and you hadn’t switched on your smartphone’s audio recorder either. Fortunately, immediately after lunch, you diligently scrambled to “download” salient details from memory into some rough consulting notes and tried to mark which points were from users and which are hypotheses or opinions not yet proven to be requirements expressed by customer users.

It will help tremendously to re-write these points you could recall into a more coherent set of user experience narratives (UXNs) for Megan, who will use this information to help tailor the look and feel of the system:

5. Information for customer persona and user experience (UX) specifications:

    5.1. Marius’ stories all described customers as “he”, which is partly understandable since most fishermen are male, although it still struck you as insensitive. But, interestingly, you’ve learnt from Marius that, although “fisher” is a term commonly used in contemporary government, scientific and conservation literature to describe people who fish for a living, female fishermen colloquially prefer “fisherman” rather than

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"fisherwoman" or "fisherperson". Marius jokingly rationalised that it made sense because of the "-man" being pronounced "min" and not "man". Related, we cannot categorise all customers as anglers because this term describes fishermen who use a rod for recreation and not a net or who fish for a living.

5.2. Some stories involved conversations about concern for poaching and dwindling fish stocks. Although most customers are hobbyist fishermen, they are not only concerned about scarce resources and severely limited permits and fishing licences for their children, but say they fear that marine resources will "run out" in their own lifetimes.

5.3. Although most customers by number are hobby fishermen, Kabeljou's majority revenue is derived from small-scale commercial fishermen and fishing companies. You asked how this market segment, distinct from the hobbyist, might use the system differently, to which Marius referred to some of his analyses of sales data to estimate that the differences are in the frequency of visits and purchases and the quantities and values per cart. This means that, because a UXN can pertain to only one persona, we will need to clearly identify which persona we're defining in each of our UXNs.

5.4. Almost all commercial fishermen have backgrounds that are similar to Anton's. (See the scenario document for Module 1.)

5.5. Marius suspects most customers would want to use the system on their smartphones and not on PCs or laptops. This is why he mentioned the need for a mobile interface in the early stages of the project. (He knows that most of his hobby peers would rather want to put up the "Gone fishing" sign than be ensconced at their desks.)

5.6. He says customers want a catalogue that is quick and easy to navigate. When you probed for what this means, Marius offered examples given by some customers that each catalogue page must show thumbnail pictures of similar or alternative products on the same page as the product being viewed so that the user can navigate to those other products' pages in one click.

5.7. To increase sales, Marius wants to encourage users to purchase immediately, so he wants a button like "Check out now" to be very prominent on every catalogue page. He said customers want to very easily see how to place their order.

5.8. He said that customers often want to be able to click a button to get more details about a product, but that the existing catalogue gives technical specifications on only a few products, not all of them.

5.9. Having specialised notes about each product's features and benefits would be appealing to customers because it would make the catalogue convenient as a one-stop source of information.

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For example, lures are highly specialised and there's a lot of scientific research on which lures work for certain fish species, to the point that different populations of the same species, separated only by their geographic location, might have different lures that work best for each population group. You shared the ironic humour of this metaphor for Kabeljou's own market segment analysis!

5.10. Customers said they would be more likely to look for more products or actually buy from Kabeljou if they could get additional product data and tips directly from the catalogue. Currently, they would visit their local fishing shop for the extra information and then end up buying the product from the shop.

5.11. Marius says that one challenge they often struggle with is to introduce new products – not new to Kabeljou, but new to the industry. The superstition and tradition amongst many fishermen is that experimenting and trying new products is accelerated when the product is successful in competition, but that an untried product will barely be noticed if it doesn't have the credibility of a big-name fisherman's success to back it up. Anton is exploring how to increase product sponsorships with top fishermen as a way of getting testimonials, although we don't know yet if testimonials on catalogue pages will make any difference for customers.

5.12. Marius says many customers are still concerned about online shopping and the safety of their credit card information. Customers said they would feel reassured if some type of one-time-password (OTP) is SMSed to them in order to complete each transaction.

5.13. Marius related a few points customers made about the overall "look and feel" of the site, like having a consistent layout and colour palette across all pages so that users can easily navigate and use each page. The current catalogue has inconsistent layouts for product pages because the different web developers, who create and manage the catalogue pages, apply their own disparate styles. Customers don't want to have to "learn" how to navigate every product page. As Marius says, "Once you've seen the layout of one page, you've seen the layout of every page."

5.14. A couple of customers suggested that the Kabeljou brand should be "subtly prominent" because it will help reinforce a sense of trust in the site, yet not be in the way. Marius coined this description in response to his irritation with the new trend of websites to have very tall top banners that use up a lot of screen space; he wants the user to be able to maximise the functional area of each page without being forced to scroll many times just to complete a task, especially for mobile users.

5.15. Customers have suggested that maybe a pop-up help box might be useful for when a user mouses over a button with only an icon and no label. But, Marius gets irritated by pop-ups when they block his sight of what's behind the pop-up, especially when he has learnt what the button will do, so he'd like some way for the user to be able to switch off this pop-up help function.

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5.16. Related to preferences for on-screen help pop-ups on buttons, you hypothesised about a few other aspects that weren't sourced from customers, like user preferences and how to have the system "remember" their preferences for subsequent visits to the site, such as whether to use imperial or metric measurements, or Celsius or Fahrenheit. You will need to research and specify other features that a typical user might want to customise later.

5.17. Marius is pleased that the new system will instantly confirm the purchase on-screen as well as the purchase details and forecast delivery date, compared with the current system that relies on manual payment reconciliation and sometimes several days' waiting between the customer making the purchase and Kabeljou emailing confirmation of the order to the customer. Customers have said that having orders confirmed immediately is a factor that could encourage them to buy from Kabeljou rather than from their local brick-and-mortar fishing shops.

Use this information for your informal meeting with Marius to summarise the UX requirements of the system.

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